

Artifiical Intelligence

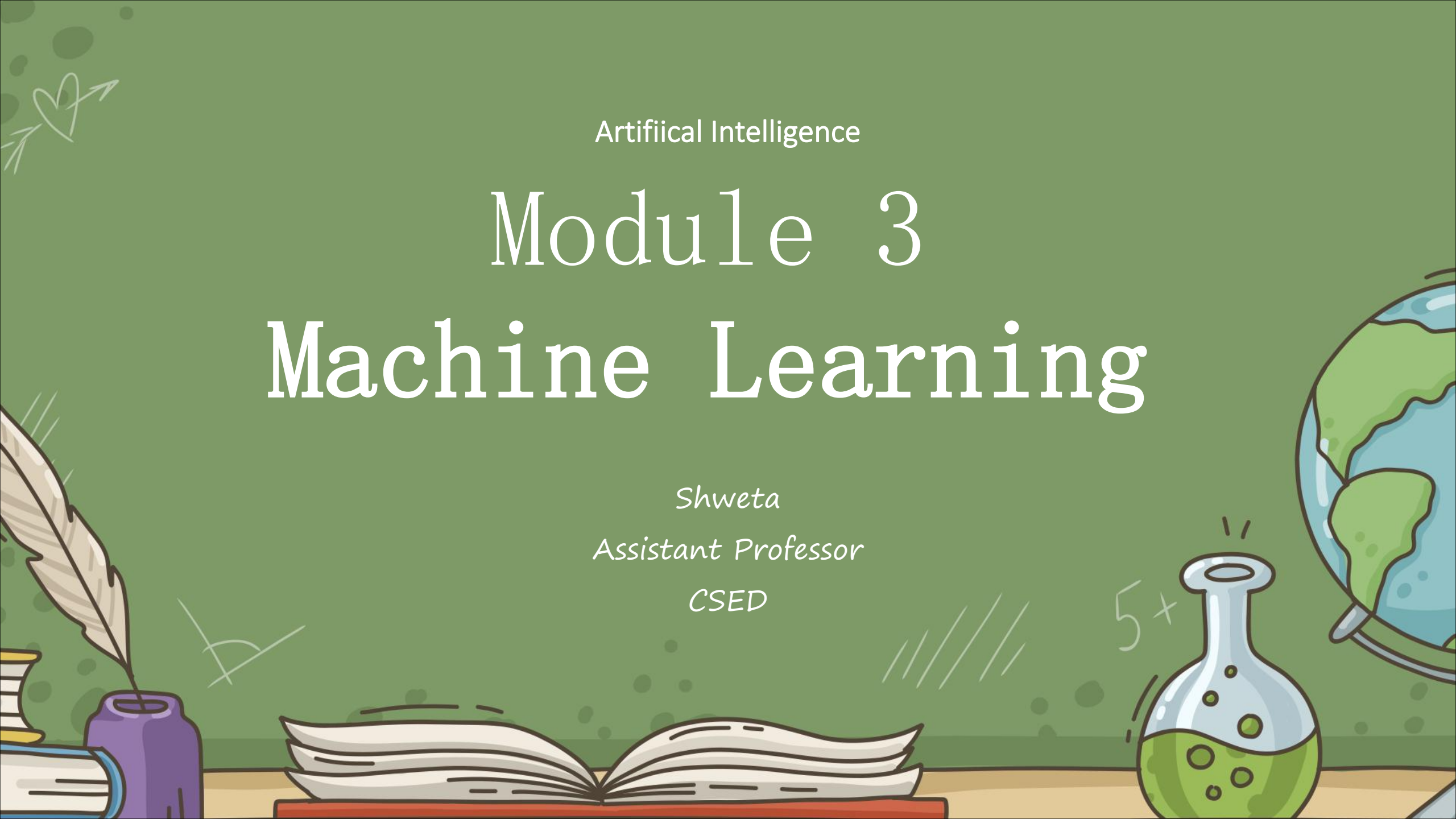
Module 3

Machine Learning

Shweta

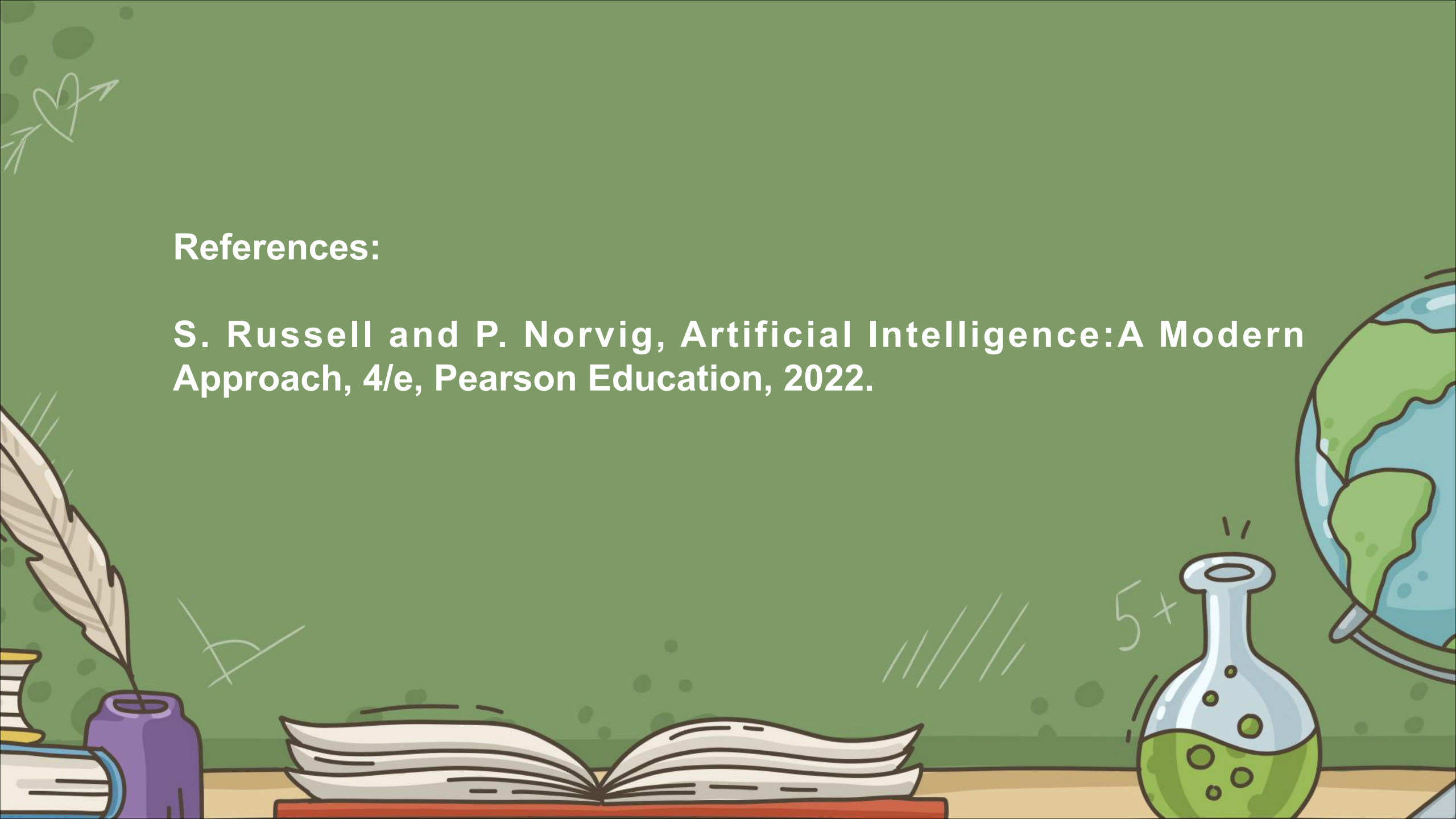
Assistant Professor

CSED

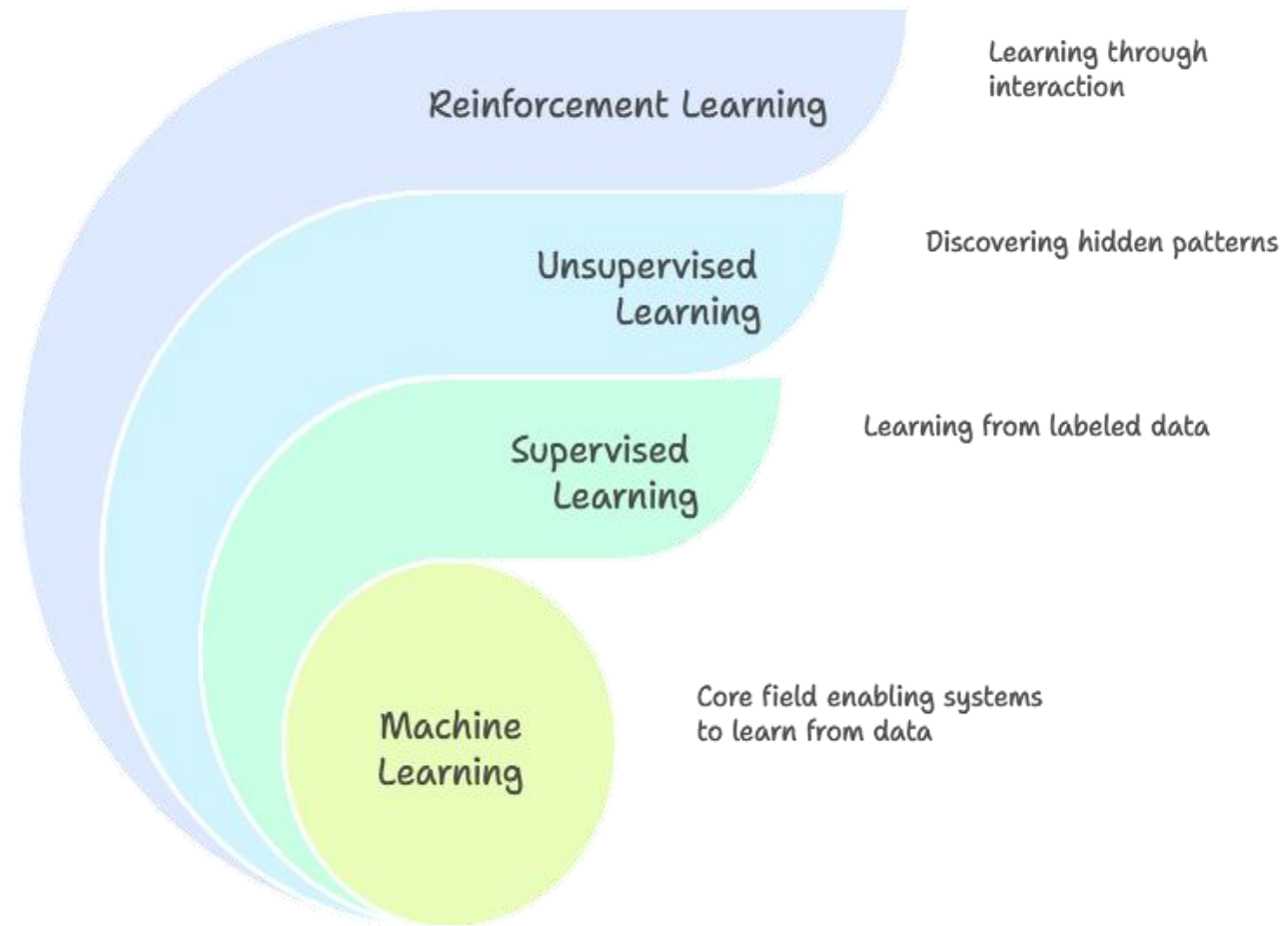


References:

S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4/e, Pearson Education, 2022.



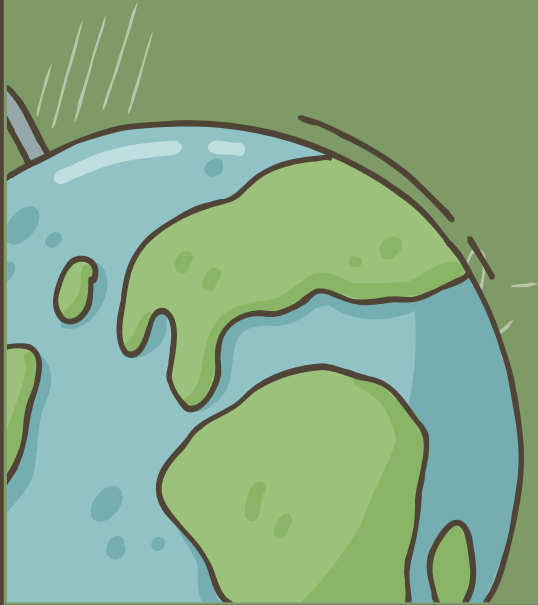
Introduction



$$A+B=C$$

Supervised Learning

$$\sqrt{a}$$



Supervised Learning

- Given a training set of N example input–output pairs

Training set $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N),$

where each pair was generated by an unknown function $y = f(x)$, discover a function h that approximates the true function f .

Supervised Learning

Goal: Build a system that predicts the price of a house based on its features.

1. Data Collection (Labeled Data) like this:

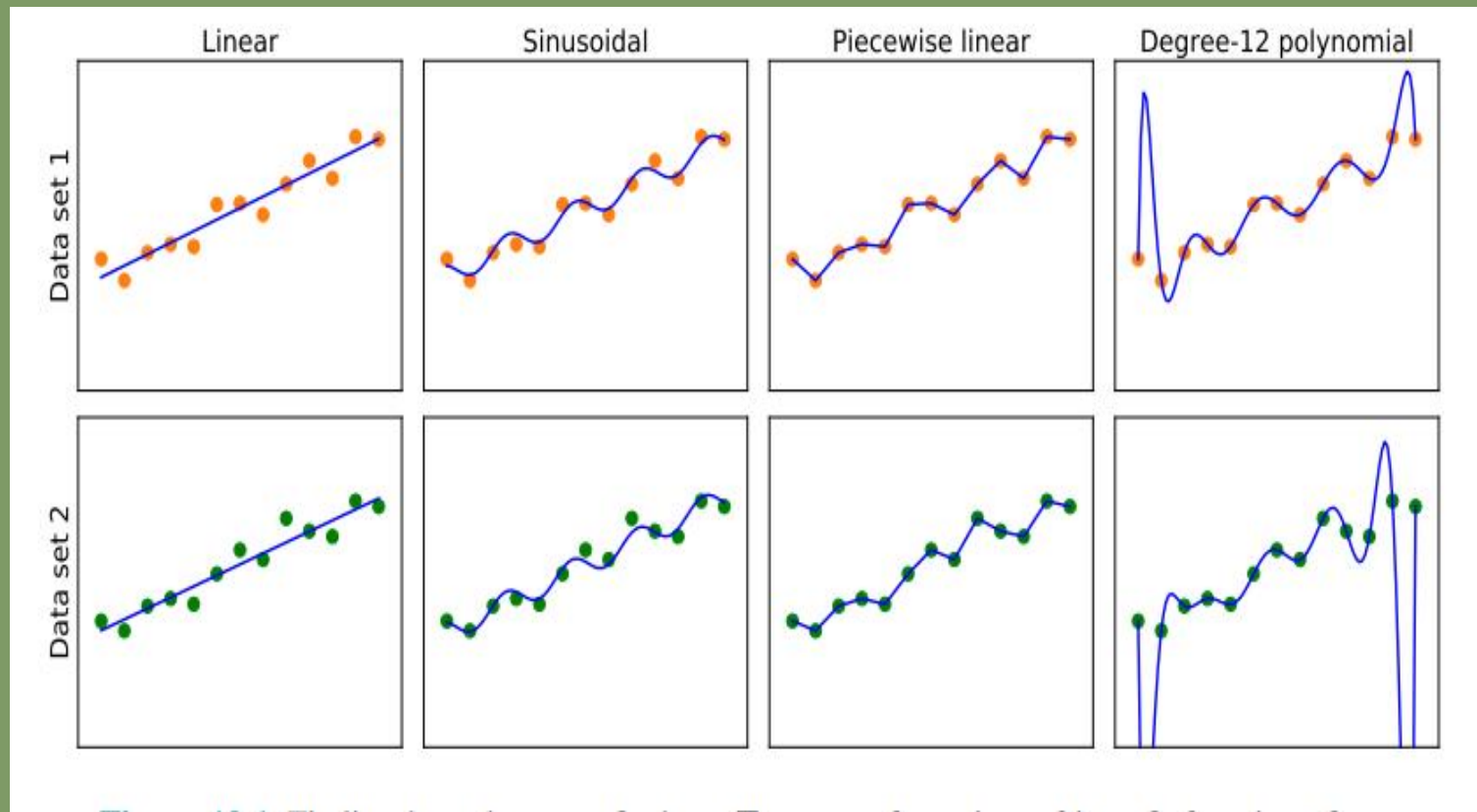
Size (sq ft)	No. of Bedrooms	Age of House (years)	Price (₹)
1000	2	10	50,00,000
1500	3	5	75,00,000
2000	4	2	1,00,00,000

Here:

Input (X) → Size, Bedrooms, Age (features)

Output (Y) → Price (label)

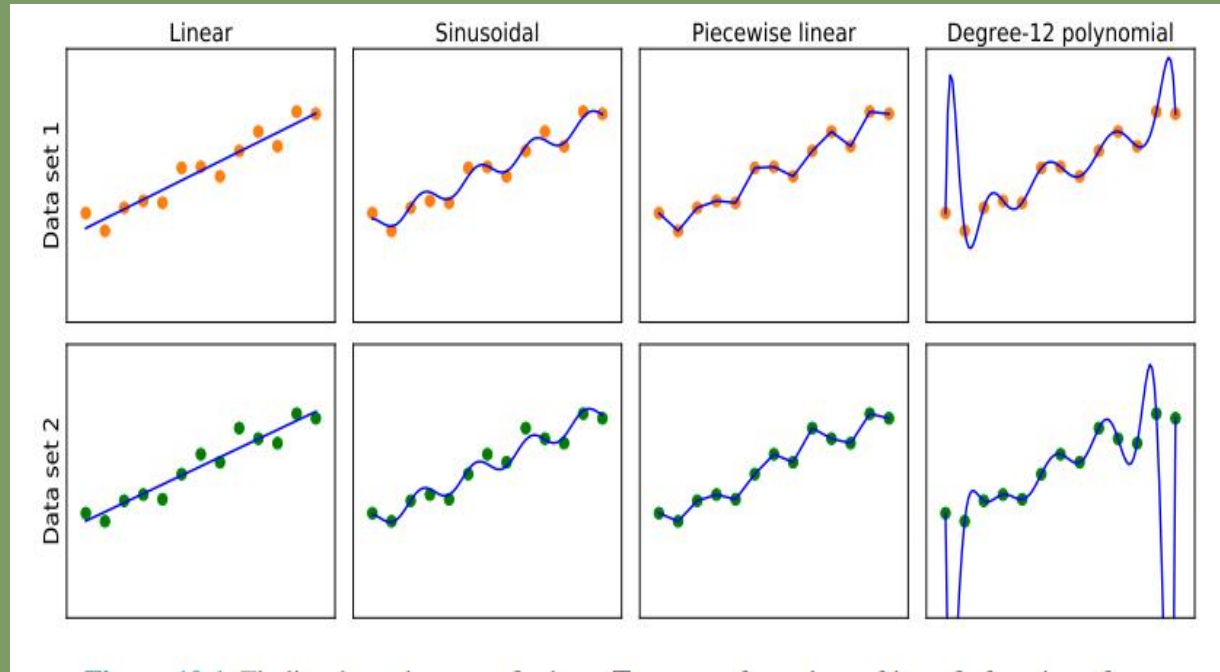
Supervised Learning



- Bias
- Variance
- bias–variance tradeoff

Finding hypotheses to fit data

Supervised Learning



Finding hypotheses to fit data

$$h^* = \operatorname{argmax}_{h \in \mathcal{H}} P(h|data).$$

By Bayes' rule this is equivalent to

$$h^* = \operatorname{argmax}_{h \in \mathcal{H}} P(data|h)P(h).$$

- expressiveness
- computational complexity

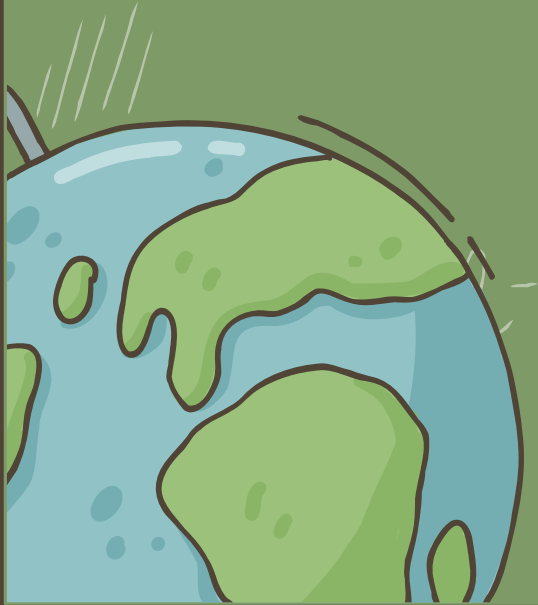
$$A+B=C$$

Supervised Learning

Decision Trees



$$\sqrt{a}$$





Restaurant waiting

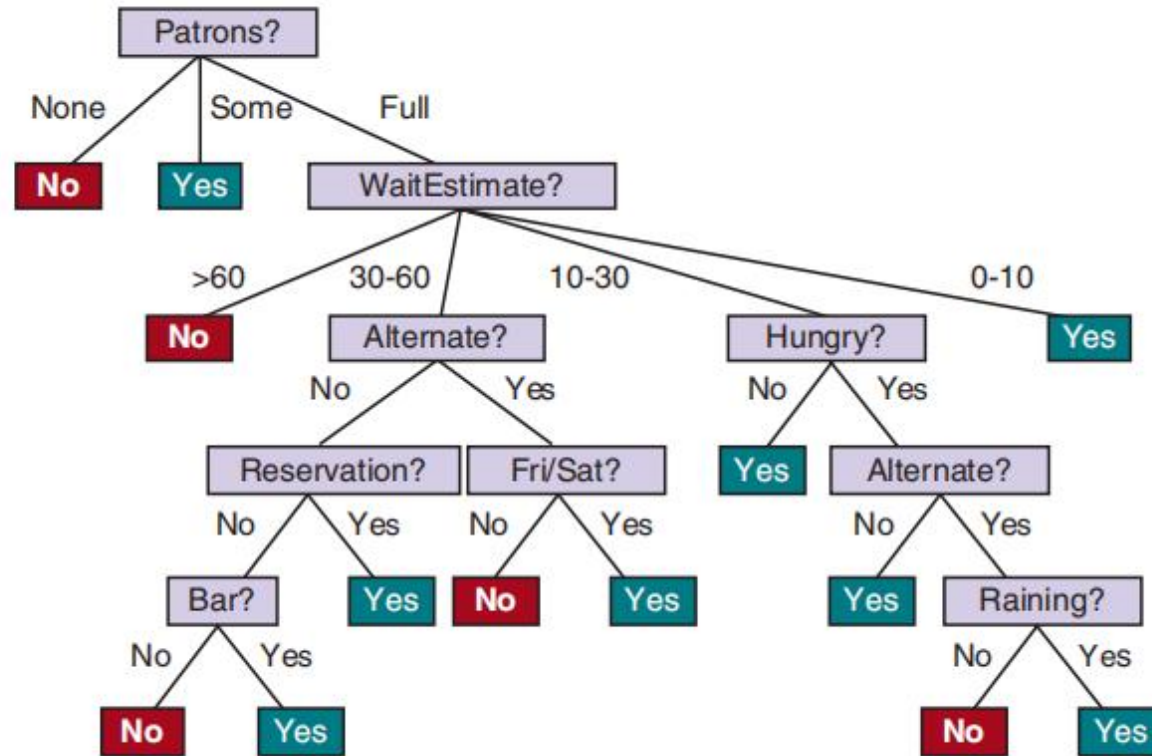
Example problem

- **Output, y ,** is a Boolean variable that we will call WillWait; it is true for examples where we do wait .
- **Input, x** vector of ten attribute values, each of which has discrete values:
 - 1. Alternate: whether there is a suitable alternative restaurant nearby.
 - 2. Bar: whether the restaurant has a comfortable bar area to wait in.
 - 3. Fri/Sat: true on Fridays and Saturdays.
 - 4. Hungry: whether we are hungry right now.
 - 5. Patrons: how many people are in the restaurant (values are None, Some, and Full).
 - 6. Price: the restaurant's price range (\$, \$\$, \$\$\$).
 - 7. Raining: whether it is raining outside.
 - 8. Reservation: whether we made a reservation.
 - 9. Type: the kind of restaurant (French, Italian, Thai, or burger).
 - 10. WaitEstimate: host's wait estimate: 0–10, 10–30, 30–60, or >60 minutes.

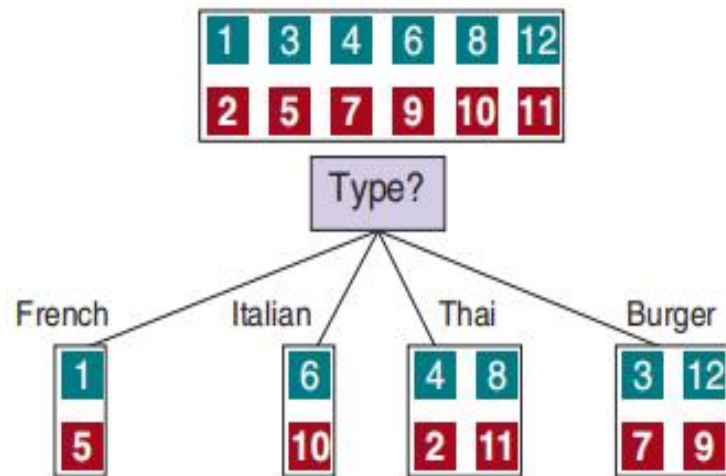


Example	Input Attributes										Output
	<i>Alt</i>	<i>Bar</i>	<i>Fri</i>	<i>Hun</i>	<i>Pat</i>	<i>Price</i>	<i>Rain</i>	<i>Res</i>	<i>Type</i>	<i>Est</i>	<i>WillWait</i>
x_1	<i>Yes</i>	<i>No</i>	<i>No</i>	<i>Yes</i>	<i>Some</i>	<i>\$\$\$</i>	<i>No</i>	<i>Yes</i>	<i>French</i>	<i>0–10</i>	$y_1 = \text{Yes}$
x_2	<i>Yes</i>	<i>No</i>	<i>No</i>	<i>Yes</i>	<i>Full</i>	<i>\$</i>	<i>No</i>	<i>No</i>	<i>Thai</i>	<i>30–60</i>	$y_2 = \text{No}$
x_3	<i>No</i>	<i>Yes</i>	<i>No</i>	<i>No</i>	<i>Some</i>	<i>\$</i>	<i>No</i>	<i>No</i>	<i>Burger</i>	<i>0–10</i>	$y_3 = \text{Yes}$
x_4	<i>Yes</i>	<i>No</i>	<i>Yes</i>	<i>Yes</i>	<i>Full</i>	<i>\$</i>	<i>Yes</i>	<i>No</i>	<i>Thai</i>	<i>10–30</i>	$y_4 = \text{Yes}$
x_5	<i>Yes</i>	<i>No</i>	<i>Yes</i>	<i>No</i>	<i>Full</i>	<i>\$\$\$</i>	<i>No</i>	<i>Yes</i>	<i>French</i>	<i>>60</i>	$y_5 = \text{No}$
x_6	<i>No</i>	<i>Yes</i>	<i>No</i>	<i>Yes</i>	<i>Some</i>	<i>\$\$</i>	<i>Yes</i>	<i>Yes</i>	<i>Italian</i>	<i>0–10</i>	$y_6 = \text{Yes}$
x_7	<i>No</i>	<i>Yes</i>	<i>No</i>	<i>No</i>	<i>None</i>	<i>\$</i>	<i>Yes</i>	<i>No</i>	<i>Burger</i>	<i>0–10</i>	$y_7 = \text{No}$
x_8	<i>No</i>	<i>No</i>	<i>No</i>	<i>Yes</i>	<i>Some</i>	<i>\$\$</i>	<i>Yes</i>	<i>Yes</i>	<i>Thai</i>	<i>0–10</i>	$y_8 = \text{Yes}$
x_9	<i>No</i>	<i>Yes</i>	<i>Yes</i>	<i>No</i>	<i>Full</i>	<i>\$</i>	<i>Yes</i>	<i>No</i>	<i>Burger</i>	<i>>60</i>	$y_9 = \text{No}$
x_{10}	<i>Yes</i>	<i>Yes</i>	<i>Yes</i>	<i>Yes</i>	<i>Full</i>	<i>\$\$\$</i>	<i>No</i>	<i>Yes</i>	<i>Italian</i>	<i>10–30</i>	$y_{10} = \text{No}$
x_{11}	<i>No</i>	<i>No</i>	<i>No</i>	<i>No</i>	<i>None</i>	<i>\$</i>	<i>No</i>	<i>No</i>	<i>Thai</i>	<i>0–10</i>	$y_{11} = \text{No}$
x_{12}	<i>Yes</i>	<i>Yes</i>	<i>Yes</i>	<i>Yes</i>	<i>Full</i>	<i>\$</i>	<i>No</i>	<i>No</i>	<i>Burger</i>	<i>30–60</i>	$y_{12} = \text{Yes}$

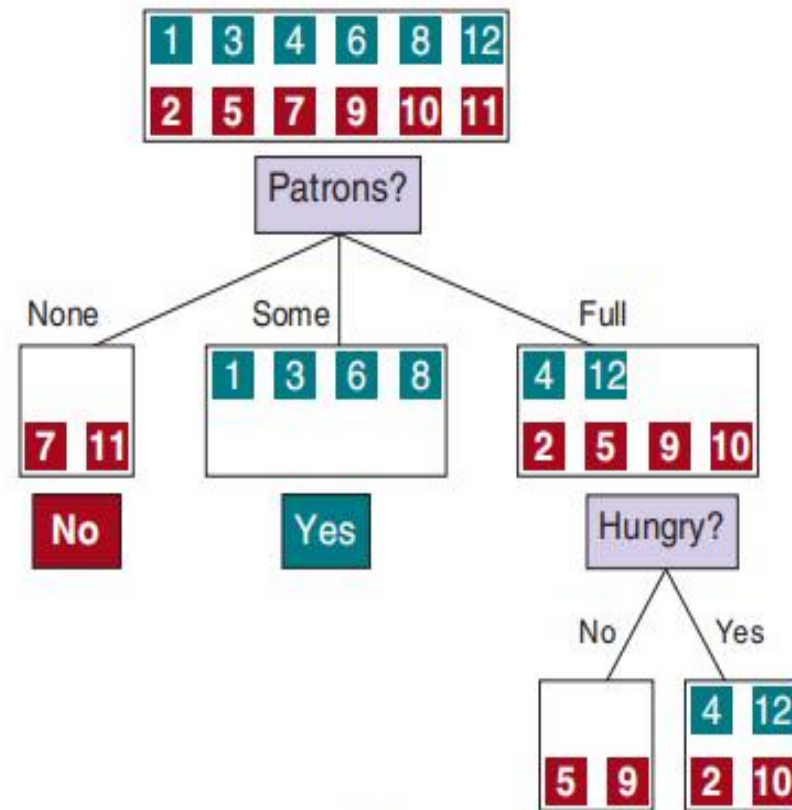
Decision Trees



Decision Trees



(a)

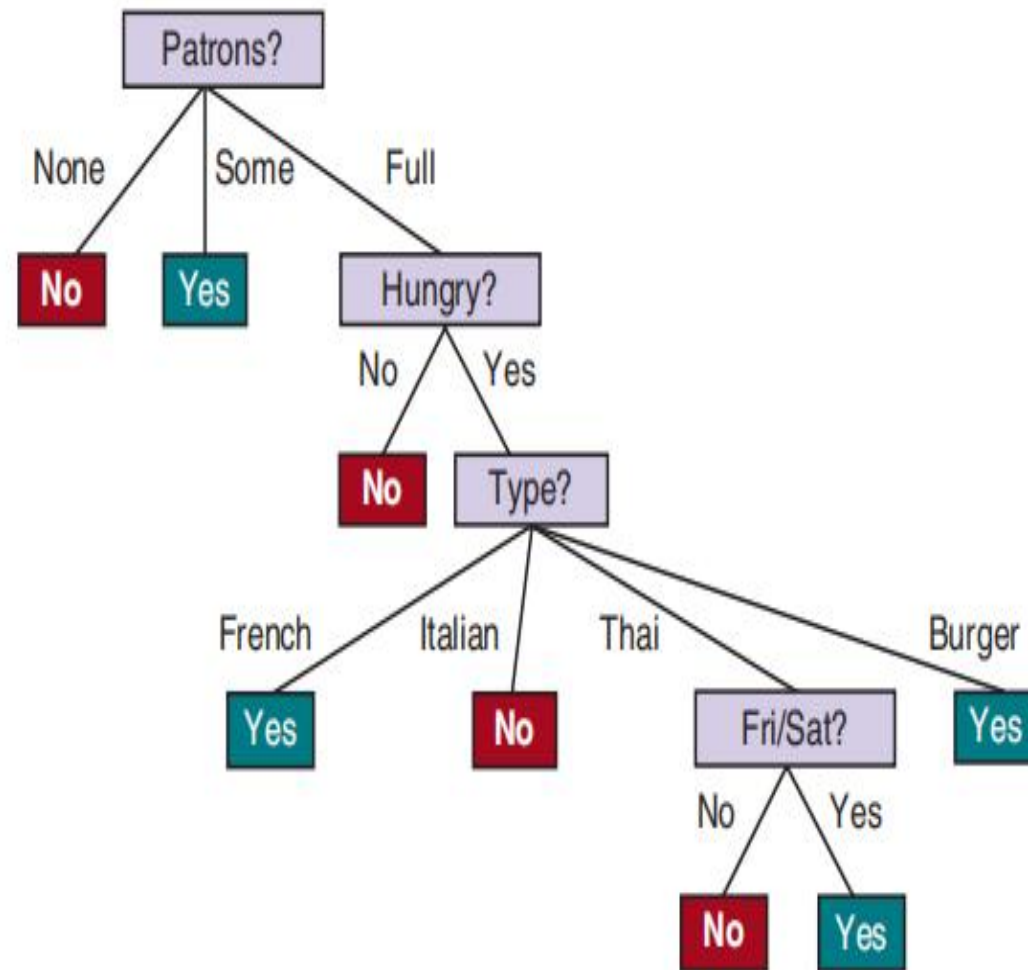


(b)

Decision Trees

```
function LEARN-DECISION-TREE(examples, attributes, parent_examples) returns a tree

  if examples is empty then return PLURALITY-VALUE(parent_examples)
  else if all examples have the same classification then return the classification
  else if attributes is empty then return PLURALITY-VALUE(examples)
  else
     $A \leftarrow \operatorname{argmax}_{a \in \text{attributes}} \text{IMPORTANCE}(a, \text{examples})$ 
    tree  $\leftarrow$  a new decision tree with root test A
    for each value v of A do
       $\text{exs} \leftarrow \{e : e \in \text{examples} \text{ and } e.A = v\}$ 
      subtree  $\leftarrow$  LEARN-DECISION-TREE(exs, attributes − A, examples)
      add a branch to tree with label (A = v) and subtree subtree
    return tree
```



Decision Trees

- **Choosing attribute tests**

- **Information Gain**
 - **Entropy**

- **Entropy:** The uncertainty of a random variable

$$\text{Entropy: } H(V) = \sum_k P(v_k) \log_2 \frac{1}{P(v_k)} = - \sum_k P(v_k) \log_2 P(v_k).$$

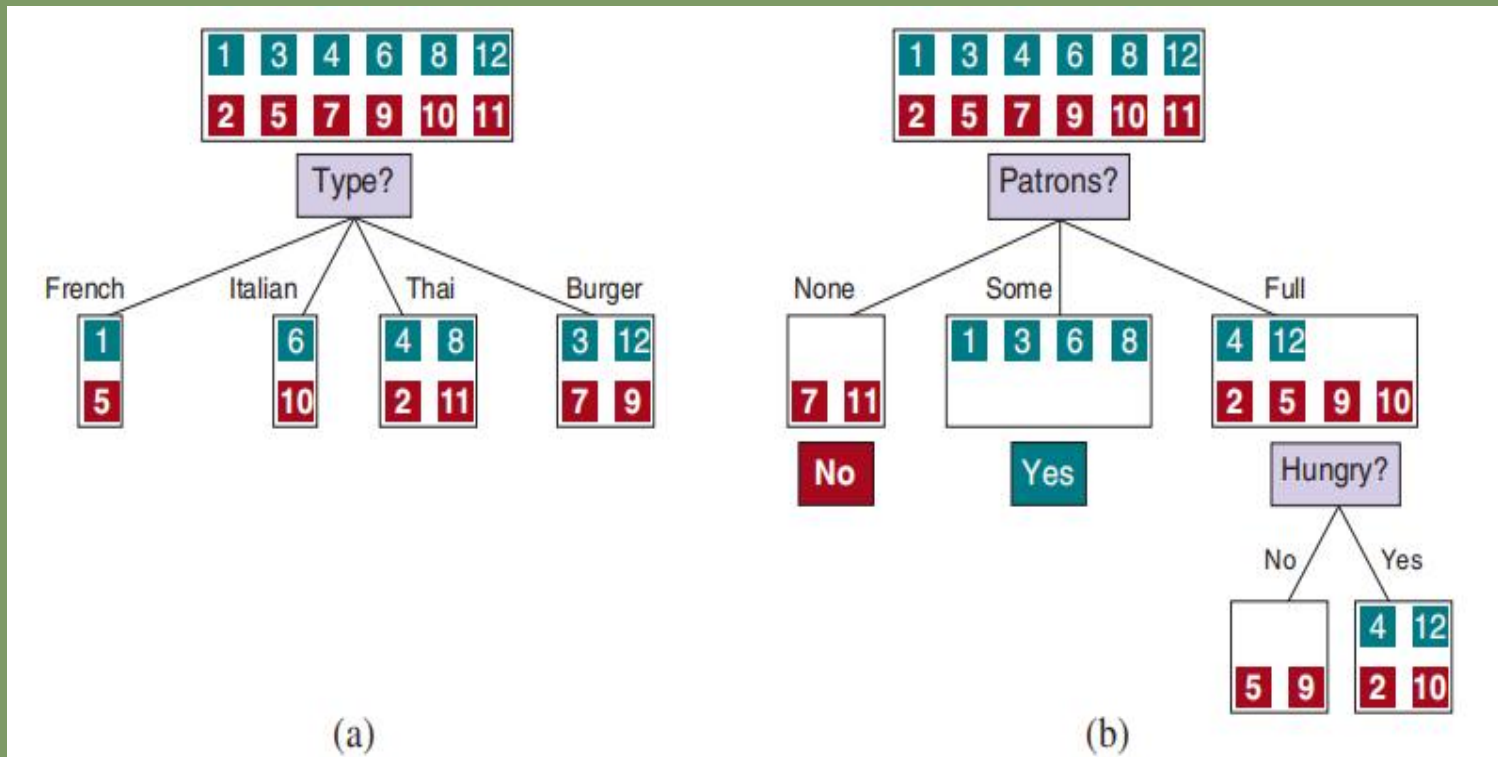
- $B(q) = - (q \log_2 q + (1-q) \log_2 (1-q))$

$$H(\text{Output}) = B(p/p+n)$$

$$\text{Remainder}(A) = \sum_{k=1}^d \frac{p_k + n_k}{p+n} B\left(\frac{p_k}{p_k + n_k}\right).$$

$$\text{Gain}(A) = B\left(\frac{p}{p+n}\right) - \text{Remainder}(A).$$

Information Gain



$$Gain(Patrons) = 1 - \left[\frac{2}{12}B\left(\frac{0}{2}\right) + \frac{4}{12}B\left(\frac{4}{4}\right) + \frac{6}{12}B\left(\frac{2}{6}\right) \right] \approx 0.541 \text{ bits},$$

$$Gain(Type) = 1 - \left[\frac{2}{12}B\left(\frac{1}{2}\right) + \frac{2}{12}B\left(\frac{1}{2}\right) + \frac{4}{12}B\left(\frac{2}{4}\right) + \frac{4}{12}B\left(\frac{2}{4}\right) \right] = 0 \text{ bits},$$

Generalization and overfitting

Decision Tree Pruning

significance testing: χ^2 pruning

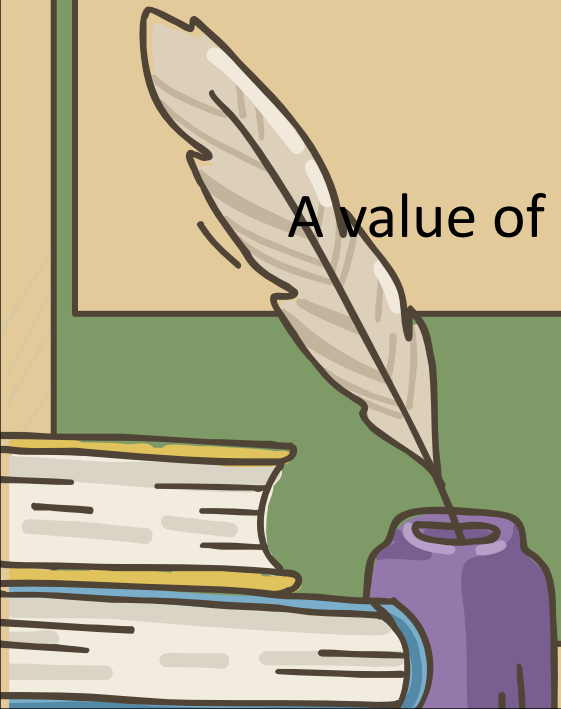
Null Hypothesis: Attribute is irrelevant

$$\hat{p}_k = p \times \frac{p_k + n_k}{p + n} \quad \hat{n}_k = n \times \frac{p_k + n_k}{p + n}.$$

A convenient measure of the total deviation is given by

$$\Delta = \sum_{k=1}^d \frac{(p_k - \hat{p}_k)^2}{\hat{p}_k} + \frac{(n_k - \hat{n}_k)^2}{\hat{n}_k}.$$

A value of $\Delta=7.82$ or more would reject the null hypothesis at the 5% level



$$A+B=C$$

Linear Regression

$$\sqrt{a}$$

$$2^{-}$$



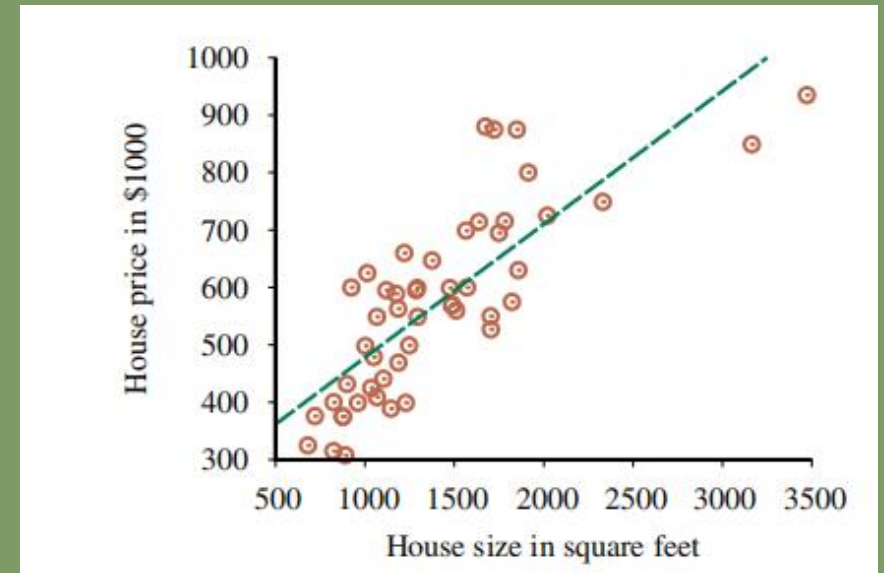


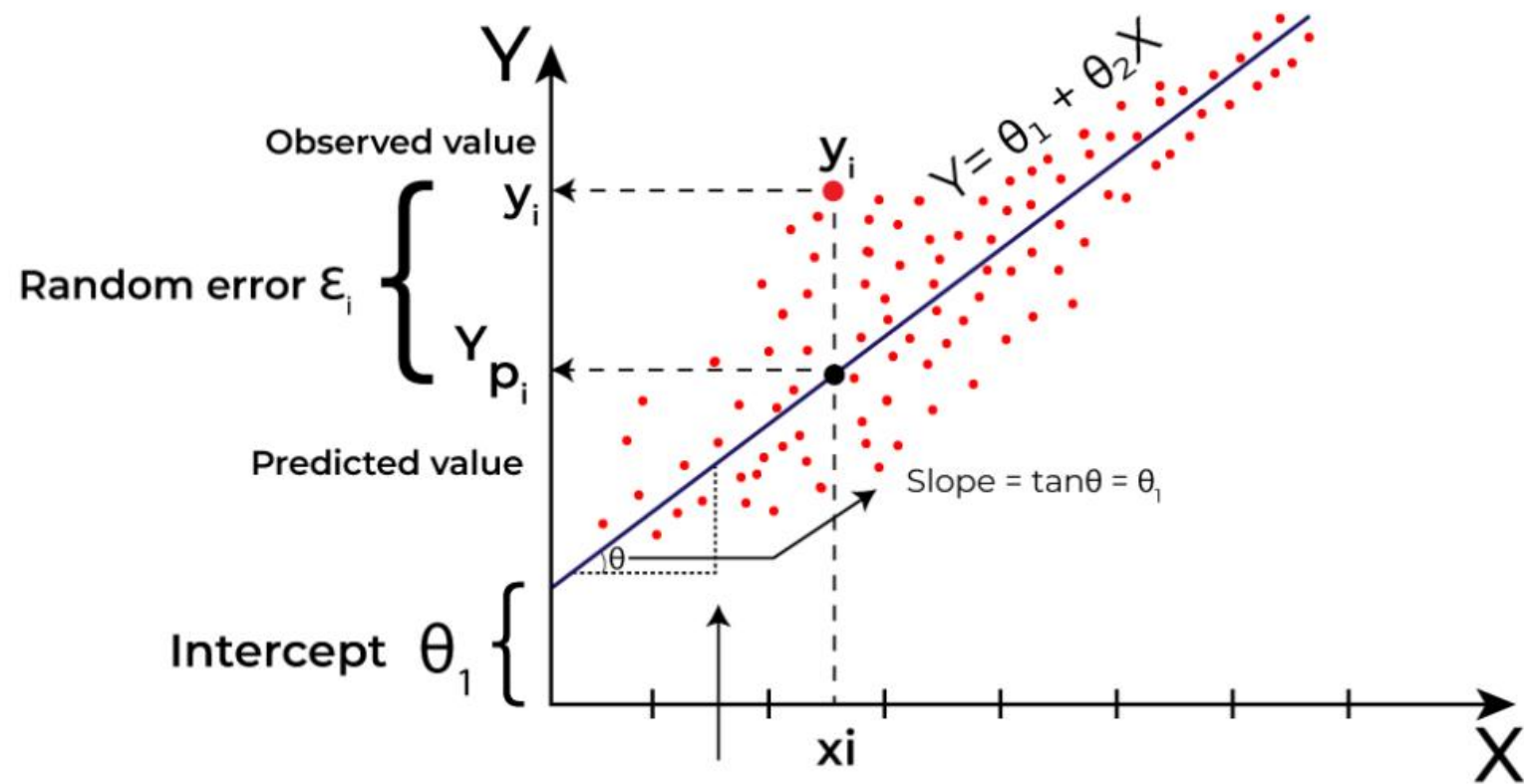
Linear Regression: Univariate Linear Function

- We'll define w to be the vector (w_0, w_1) , and define the linear function with those weights as :

$$h_w(x) = w_1x + w_0$$

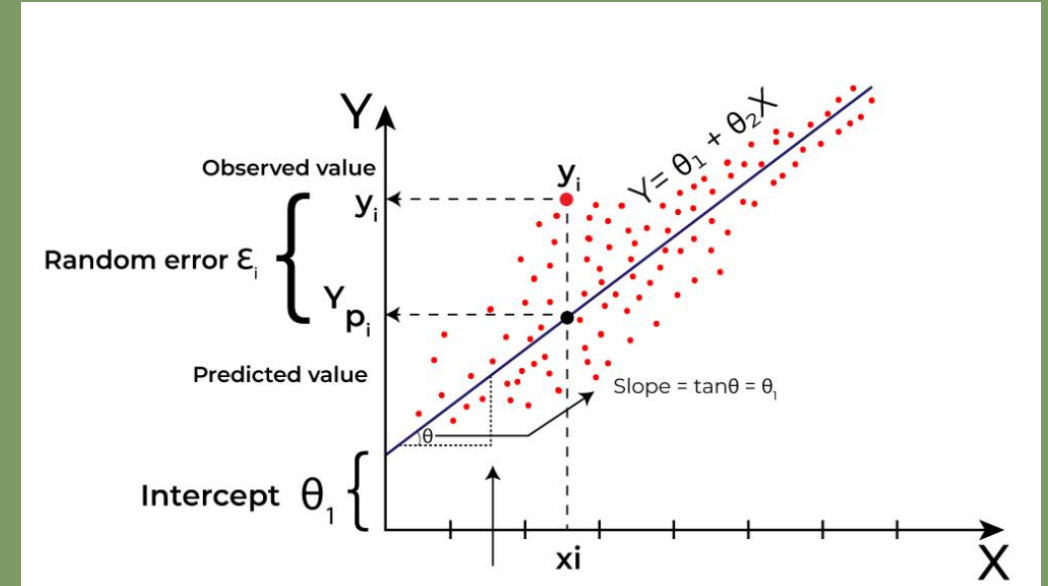
- The task of finding the h_w that best fits these data is called linear regression.





squared-error loss function, L_2

$$Loss(h_{\mathbf{w}}) = \sum_{j=1}^N L_2(y_j, h_{\mathbf{w}}(x_j)) = \sum_{j=1}^N (y_j - h_{\mathbf{w}}(x_j))^2 = \sum_{j=1}^N (y_j - (w_1 x_j + w_0))^2.$$



$$\frac{\partial}{\partial w_0} \sum_{j=1}^N (y_j - (w_1 x_j + w_0))^2 = 0 \text{ and } \frac{\partial}{\partial w_1} \sum_{j=1}^N (y_j - (w_1 x_j + w_0))^2 = 0.$$

Gradient descent

The algorithm is as follows:

$\mathbf{w} \leftarrow$ any point in the parameter space

while not converged do

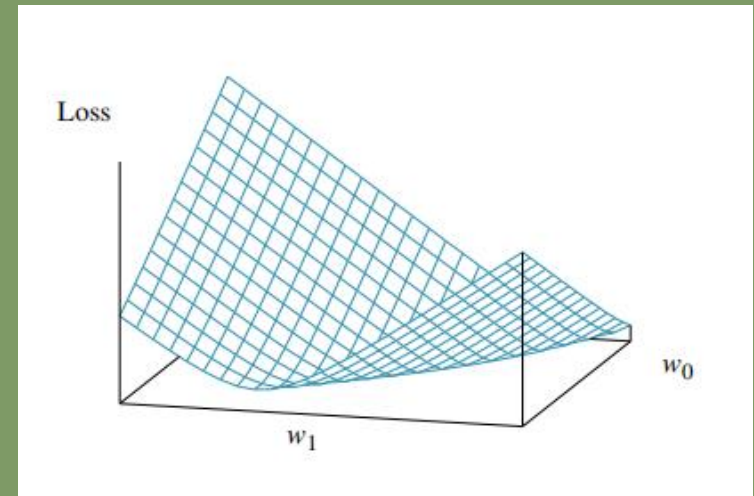
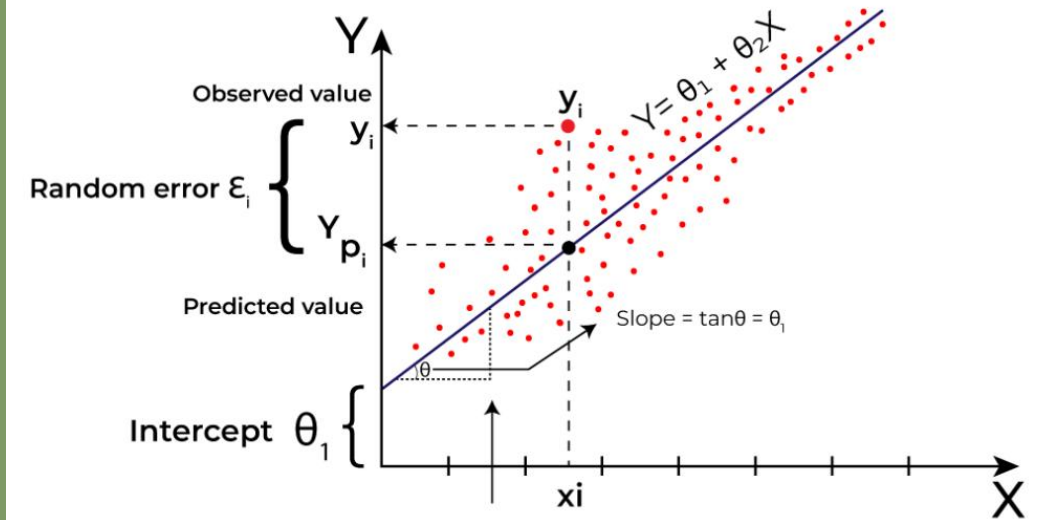
for each w_i in $\mathbf{w}$ do

$$w_i \leftarrow w_i - \alpha \frac{\partial}{\partial w_i} \text{Loss}(\mathbf{w})$$

$$\text{Loss}(h_{\mathbf{w}}) = \sum_{j=1}^N L_2(y_j, h_{\mathbf{w}}(x_j)) = \sum_{j=1}^N (y_j - h_{\mathbf{w}}(x_j))^2 = \sum_{j=1}^N (y_j - (w_1 x_j + w_0))^2$$

$$\frac{\partial}{\partial w_0} \sum_{j=1}^N (y_j - (w_1 x_j + w_0))^2 = 0 \text{ and } \frac{\partial}{\partial w_1} \sum_{j=1}^N (y_j - (w_1 x_j + w_0))^2 = 0$$

$$\frac{\partial}{\partial w_0} \text{Loss}(\mathbf{w}) = -2(y - h_{\mathbf{w}}(x)); \quad \frac{\partial}{\partial w_1} \text{Loss}(\mathbf{w}) = -2(y - h_{\mathbf{w}}(x)) \times x$$



Gradient descent

The algorithm is as follows:

$\mathbf{w} \leftarrow$ any point in the parameter space

while not converged do

for each w_i in $\mathbf{w}$ do

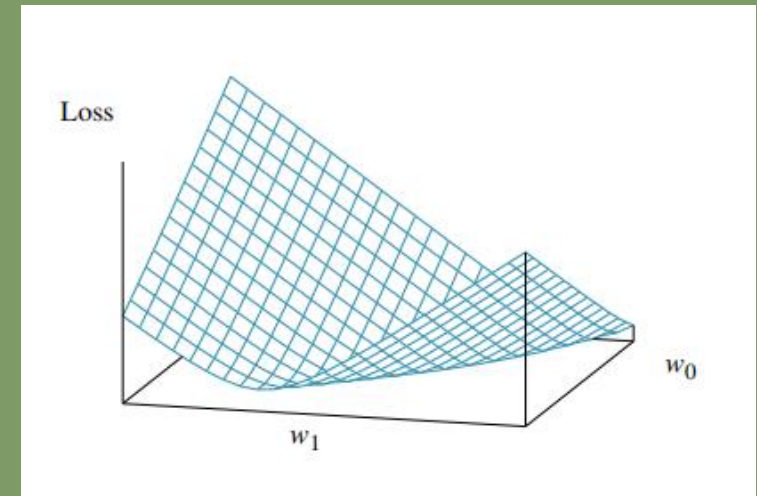
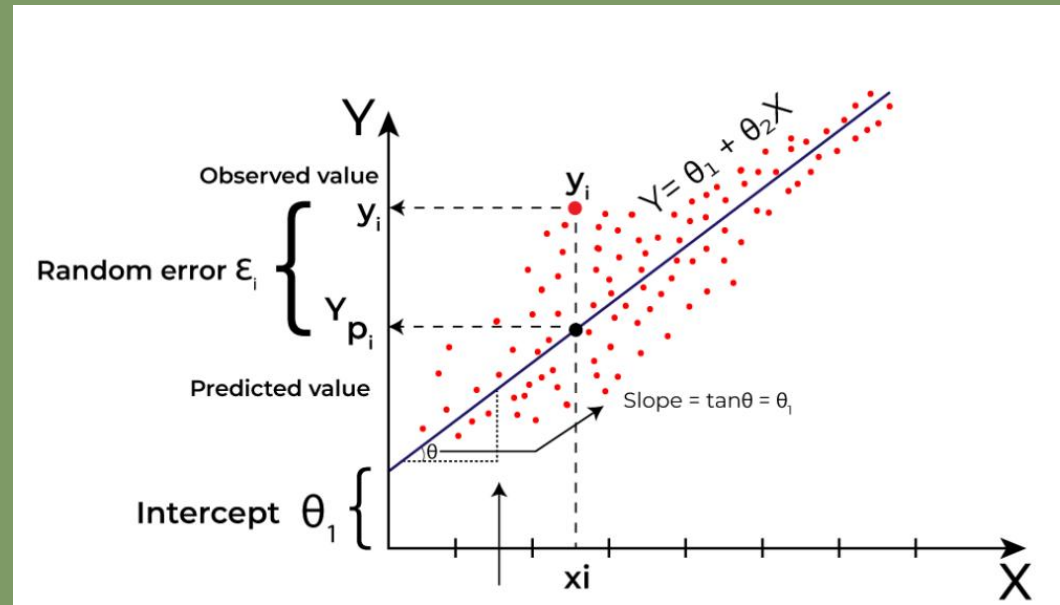
$$w_i \leftarrow w_i - \alpha \frac{\partial}{\partial w_i} \text{Loss}(\mathbf{w})$$

$$\text{Loss}(h_{\mathbf{w}}) = \sum_{j=1}^N L_2(y_j, h_{\mathbf{w}}(x_j)) = \sum_{j=1}^N (y_j - h_{\mathbf{w}}(x_j))^2 = \sum_{j=1}^N (y_j - (w_1 x_j + w_0))^2$$

$$\frac{\partial}{\partial w_0} \sum_{j=1}^N (y_j - (w_1 x_j + w_0))^2 = 0 \text{ and } \frac{\partial}{\partial w_1} \sum_{j=1}^N (y_j - (w_1 x_j + w_0))^2 = 0$$

$$\frac{\partial}{\partial w_0} \text{Loss}(\mathbf{w}) = -2(y - h_{\mathbf{w}}(x)); \quad \frac{\partial}{\partial w_1} \text{Loss}(\mathbf{w}) = -2(y - h_{\mathbf{w}}(x)) \times x$$

$$w_0 \leftarrow w_0 + \alpha (y - h_{\mathbf{w}}(x)); \quad w_1 \leftarrow w_1 + \alpha (y - h_{\mathbf{w}}(x)) \times x$$



$$A+B=C$$

Logistic Regression

$$\sqrt{a}$$

$$2^{-}$$

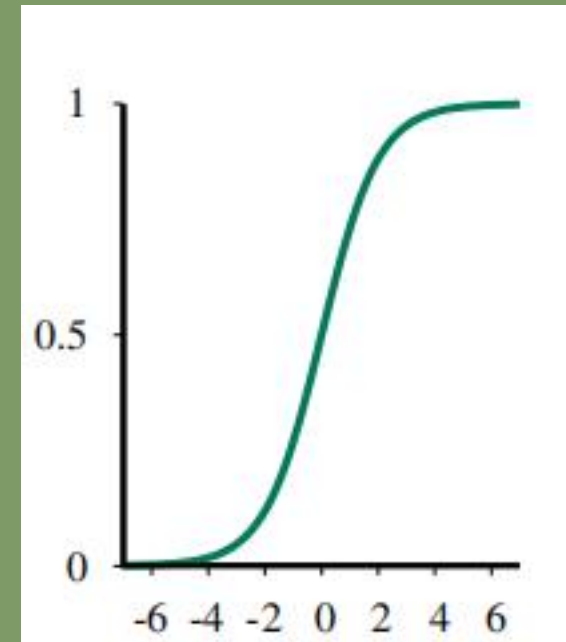


Logistic Regression

$$h_{\mathbf{w}}(\mathbf{x}) = \text{Logistic}(\mathbf{w} \cdot \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w} \cdot \mathbf{x}}}.$$

$$\begin{aligned} \frac{\partial}{\partial w_i} \text{Loss}(\mathbf{w}) &= \frac{\partial}{\partial w_i} (y - h_{\mathbf{w}}(\mathbf{x}))^2 \\ &= 2(y - h_{\mathbf{w}}(\mathbf{x})) \times \frac{\partial}{\partial w_i} (y - h_{\mathbf{w}}(\mathbf{x})) \\ &= -2(y - h_{\mathbf{w}}(\mathbf{x})) \times g'(\mathbf{w} \cdot \mathbf{x}) \times \frac{\partial}{\partial w_i} \mathbf{w} \cdot \mathbf{x} \\ &= -2(y - h_{\mathbf{w}}(\mathbf{x})) \times g'(\mathbf{w} \cdot \mathbf{x}) \times x_i. \end{aligned}$$

where g is representing logistic function (SIGMOID function) and g' is representing derivative of g .



$$\text{Logistic}(z) = \frac{1}{1 + e^{-z}}$$

$$g'(z) = g(z)(1 - g(z))$$

$$g'(\mathbf{w} \cdot \mathbf{x}) = g(\mathbf{w} \cdot \mathbf{x})(1 - g(\mathbf{w} \cdot \mathbf{x})) = h_{\mathbf{w}}(\mathbf{x})(1 - h_{\mathbf{w}}(\mathbf{x}))$$

Learning rule:

$$w_i \leftarrow w_i + \alpha (y - h_{\mathbf{w}}(\mathbf{x})) \times h_{\mathbf{w}}(\mathbf{x})(1 - h_{\mathbf{w}}(\mathbf{x})) \times x_i.$$

